REVENG Report

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1. Introduction

The product examined in this report was a low-cost hand-crank torch identified as the DigiFlex Hand Press Torch. The aim of the work was to reverse engineer the product and develop a redesign aligned with the ethos of Lisa's Eco-Innovations. This company was selected because the main opportunities identified in the torch related to sustainability, material efficiency, and product life extension.

To support this, the torch was tested, dismantled, and analysed through a bill of materials, material and manufacturing assessment, and Eco Audit. These findings were then used to identify the most significant redesign opportunities and justify a revised design proposal for a more environmentally focused client.

2. Background

2.1 Product Category Overview

Electric torches use light-emitting diodes (LEDs) to provide illumination for everyday tasks, outdoor activities such as hiking and camping, and emergency situations. Product quality, durability and performance typically vary according to cost and target market, with most torches being powered by either rechargeable or single-use batteries.

2.2 Overview of Selected Product

The torch analysed in this report was identified as the DigiFlex Hand Press Torch, which incorporated a hand-crank dynamo system for manual electricity generation. However, visually similar products were also found on Chinese online marketplaces, suggesting that the torch may be a white-label or rebadged mass-manufactured product rather than a uniquely engineered design.

2.3 Price Range for Similar Products

Electric torch prices vary widely, from very low-cost budget products (<£0.50) to premium professional models (>£300). However, hand-crank or dynamo-powered torches are generally positioned at the lower-cost end of the market. Figure 1 shows a selection of comparable products across this range.



高品质透明双灯手压电筒手电筒

£0.17

(E.tb.cn, 2021)



£6.99

DIGIFLEX Hand Press Torch

(Amzn.eu, 2025)



£7.99

FORCLAZ DYNAMO 100

(Decathlon.co.uk, 2026a)



£19.99

FORCLAZ DYNAMO 900 PWB

(Decathlon.co.uk, 2026b)

Figure 1: Selected product (second image) and comparable products identified online.

2.4 Materials and Manufacturing Methods Found in Similar Products

Comparable hand-crank torches typically consist of a polymer outer casing, LED lighting units, a rechargeable energy-storage element, and a manually operated charging mechanism (VARTA AG, 2026; Decathlon, 2026a; Decathlon, 2026b; CN202091803U, 2011; US9212794B2, 2012). Official product pages for comparable models show that thermoplastics are commonly used for the housing, with the VARTA Dynamo Light listing ABS, PC and TPR polymers, and the Decathlon DYNAMO 100 and DYNAMO 500 listing ABS as the frame material (VARTA AG, 2026; Decathlon, 2026a; Decathlon, 2026b). Patent descriptions of similar products also identify the internal structures of a crank handle, gear transmission, generator, battery pack, and LED light source (CN202091803U, 2011; US9212794B2, 2012).

Most polymer components were likely injection moulded. This included the outer casing, light housing and smaller internal polymer components, whose geometry suggested suitability for low-cost mass production (Protolabs, 2024). The gear train components also appeared to be injection moulded polymer parts, allowing low-cost production of repeatable small features. (US20040156193A1, 2003) The metal components were geometrically simple and likely formed rather than machined, with the cross stands appearing to be stamped from sheets, the spring formed from a wire, and the hollow shaft functioning as a formed fastening feature. (Protolabs, n.d.) The electrical elements, including the coil, LEDs and switch, were likely mass-manufactured separately and assembled into the torch during final production via soldering and mounting. Together, these methods suggested that the product was optimised for low-cost mass manufacture and straightforward assembly.

3. Method

3.1 Product Testing and Evaluation

The torch was received fully packaged in a box, with a thin plastic bag inside protecting the product from surface scratches. As an initial step, available information on the packaging and manual, alongside the online description of the torch found on Amazon, was reviewed in order to build a clearer understanding of the product and its intended use.

The torch was then tested in both the hand-crank and battery-powered modes. This was done to evaluate its functionality, user interaction, and any aspects of the user experience that could be improved.

Positive aspects identified in the tests:

- The product was small and lightweight
- The product was inexpensive
- The product produced a relatively strong light output when battery-powered
- The product included a convenient wrist strap for easier holding while walking

Negative aspects identified in the tests:

- The hand-crank function generated little electricity, resulting in negligible light output
- The product did not provide an easy method of battery replacement

3.2 Product Teardown

A full teardown of the torch was carried out to separate the product into its lowest practical component level to allow for an accurate eco-audit and provide a clearer understanding of the product's materials and manufacturing processes. Throughout the teardown, photographs were taken at key stages and observations were recorded to provide better documentation of the process.

Disassembly began by removing the external casing with a Phillips screwdriver. The internal components were then detached manually from the casing where possible. Some components, such as the gear elements, were tightly retained on their shafts and required pliers for removal. Components connected by soldered joints were separated at the soldered interfaces. In total, the torch was disassembled into 28 different component types.

The dynamo was particularly difficult to remove because the rivet and mounting shaft on the casing had very little clearance. As a result, part of the casing structure had to be broken in order to extract the dynamo and disassemble it further into smaller constituent parts.

Each part was weighed individually using a digital scale with a precision of 0.01g, named, and its method of removal was logged. The components were then laid out on the display board for photographing.

3.3 Material Identification

Using information from the teardown, the materials of each component were identified. The torch was found to consist primarily of polymers and metals, with minor electrochemical sub-assemblies including the cells and LEDs.

To identify polymers, the following tests were conducted with small samples cut from each component using the plastics identification chart. (cite here appendix here)

- Using a soldering iron to observe material behaviour under heat
- A float test in water to assess relative density
- Burning a corner of the sample to observe smoke, speed of burning, flame colour, and whether the material was self-extinguishing

Burn testing indicated that the blue casing parts and cell holder were likely polypropylene, whereas the remaining polymer components, including the gears, handle parts, lens and light housing, could not be identified with certainty and were therefore assigned as likely polymer families.

Metallic components were identified by visual examination and checked with a magnet to distinguish likely ferrous parts from non-ferrous ones. Online sources were also consulted to distinguish which specific materials within each broader material family were most likely to have been used in the components. The screws, pins, spring, stamped stands, and hollow rivet shaft were all consistent with steel components, with the spring identified separately as spring steel due to its function and required fatigue resistance. The coil was clearly a non-ferrous component and was identified as a copper wire. The dynamo also contained a ring magnet, which was likely ferrite.

4. Product analysis

4.2 Bill of Materials

A bill of materials was created following teardown and material identification of the product. Table 1 shows the key components selected from the full bill of materials on the basis of functional importance, environmental impact, and redesign relevance. The full bill of materials can be found in Appendix B. The certainty of each material and manufacturing-method identification was indicated using colour coding: red for tentative, amber for probable, and green for verified.

Number	Name	Quantity	Mass per unit	Total Mass	Material	Manufacturing method	Embodied Energy	CO_2 Footprint
		(unit)	(g)	(kg)			(MJ)	(kg)
Exterior Body Components								
2	Blue Case A	1	7.25	0.00725	PP (polypropylene)	Injection moulding	0.493	0.02077125
6	Blue Case B	1	8.11	0.00811	PP (polypropylene)	Injection moulding	0.55148	0.02323515
Main White Gear Section								
7	Main White Gear	1	2.43	0.00243	POM (acetal)	Injection moulding	0.20898	0.007776
Handle Section								
9	Grey Handle Casing	1	3.44	0.00344	ABS	Injection moulding	0.321984	0.0123668
10	Black Inner Gear Handle	1	6	0.006	PA (nylon)	Injection moulding	0.861	0.04227
Dynamo Section								
15	Black Magnet Casing	1	2.36	0.00236	ABS	Injection moulding	0.220896	0.0084842
16	O Shaped Magnet	1	6.42	0.00642	Ferrite permanent magnet (cast iron)	Pressed/sintered + magnetised	0.208008	0.0152475
19	Copper Coil	1	0.78	0.00078	Enamelled copper wire	Coil winding	0.046098	*0.0028275*
Lighting Unit Section								
23	LR1130 Button Cells	3	1.1	0.0033	Alkaline button cell (steel shell + Zn/MnO2)	Purchased battery cell	0.10164	0.0076725
25	3 Through-hole LED on PCB	1	1.33	0.00133	Mixed PCB/electronic assembly	PCB assembly + through-hole soldering	0.1341305	0.00801325
26	Silver-Coated Light Housing	1	1.92	0.00192	ABS	Injection moulding + vacuum metallising	0.179712	0.0069024
28	Grey Outer Light Casing	1	2.76	0.00276	PP (polypropylene)	Injection moulding	0.18768	0.0079074

Table 1: Condensed Bill of Materials

4.3 Key Material Findings

The torch was composed primarily of polymers and metals, with smaller electrochemical sub-assemblies such as the button cells and LEDs as mentioned previously. The main polymer by mass was **PP**, as it was used for all the outer casing components. Other significant polymers included **ABS**, **PA (nylon)**, **and POM (acetal)**, which were used in smaller structural parts. The main metallic materials used were **steel**, **copper**, and **ceramic ferrite**. Overall, the certainty of the material identifications varied, with some materials verified more confidently than others.

4.3.1 PP (polypropylene)

PP was identified in both the blue casing halves and the grey outer light casing. These components were the largest components by mass; hence PP holds a significant share of the torch's structural material, making up **32.7% of the product's total mass**. These large PP parts also contributed significantly to the embodied energy **(31.5%)** and CO2 footprint **(27.7%)** of the whole product.

The confidence in PP identification was **moderate**, based on burn testing, float testing, and the geometry of the parts.

4.3.2 PA (nylon), ABS, and POM (acetal)

PA, ABS, and POM were identified in most of the other polymer-made components. Although components with PA hold only **10.6% of the total mass**, they contribute disproportionately to **21.5% of the total embodied energy** and **22.1% of the total CO2 footprint**. Parts with ABS and POM add up to **20.3% of total mass**, **26.3% of embodied energy**, and **20.9% of total CO2 footprint.**

The identification confidence for these materials was **lower than for PP**, as several internal polymers could only be identified to likely family levels rather than exact polymer grades on the basis of test results.

4.3.3 Steel

Steel was used for the main load-bearing and retention components, which are generally small but functionally important. The steel parts did not dominate the Eco Audit in the same way as polymers, contributing to **18.5% of the total mass**, but only **8.2% of total embodied energy and 12.8% of total CO2 footprint**, suggesting that steel was not the primary environmental driver of the product.

The confidence in steel identifications was **high**, based on magnetic response, visual examination, and typical use in low-cost consumer mechanisms.

4.3.4 Ferrite

The ferrite ring magnet accounted for **11.3% of total mass, 5.2% of total embodied energy, and 8.0% of total CO2 footprint**, making it the most significant non-polymer contributor in the product assembly. As permanent magnetic performance was essential to the dynamo mechanism, and ferrite is a common low-cost magnet material, it appeared to be an appropriate choice for this product.

The identification confidence in ferrite was **high**, based on magnetic response, visual examination, and typical use in low-cost consumer mechanisms.

4.3.5 Copper

Copper was used in the dynamo coil, where electrical conductivity was required for current generation. Although its total mass and environmental contribution were relatively small compared with the major polymer components, it remained functionally essential to the operation of the product.

Confidence in the copper identification was **high**, based on its visual appearance and functional role within the generator assembly.

Embodied Energy and CO2 Footprint

4.3.1 Eco Audit method and assumptions

4.3.2 Whole-product results

4.3.3 Key contributors / critical components

4.3.4 Implications for redesign